CS471 – Lesson 12 Solutions Key

Markov Decision Processes II: Values and Policies

Part 1: Guided In-Class Worksheet (Instructor-Led)

1. Policies

Scenario: An autonomous Space Force satellite is tasked with monitoring a designated sector.

a. **Question:** Circle the best definition of a policy (π):

A. A rule that maps each state to an action or action distribution.
B. A list of all rewards received during one episode.
C. A model that gives the probability of every next state.
D. A function that stores only immediate rewards.

> Solution
>
> **A. A rule that maps each state to an action or action distribution.**[cite: 9]

b. **Question:** Define a **deterministic policy** versus a **stochastic policy**.

> Solution
>
> A deterministic policy selects exactly one action per state[cite: 9, 10]. A stochastic policy assigns action probabilities (an action distribution) for each state[cite: 10].

2. State Value and Action Value

Question: Complete each mathematical statement below using $V^\pi(s)$ or $Q^\pi(s, a)$.

> Solution
>
> - Expected return starting in state s and then following policy π: $\mathbf{V}^\pi(\mathbf{s})$[cite: 9]
> - Expected return after taking action a in state s and then following policy π: $\mathbf{Q}^\pi(\mathbf{s}, \mathbf{a})$[cite: 9]

Question: Which of these two quantities compares different actions available in the same state?

> Solution
>
> The Action-Value function, $\mathbf{Q}^\pi(\mathbf{s}, \mathbf{a})$, compares different actions available in the same state[cite: 9].

Question: What does an **optimal state-value function** ($V^*(s)$) represent?

Solution

It represents the maximum expected discounted return achievable from each state[cite: 10].

3. Discounted Return (G)

Scenario: The Space Force satellite receives an immediate reward of +2 now for re-positioning, and a reward of +6 one step later for successfully taking an image. Let the discount factor $\gamma = 0.5$.

Question: Write the two-step expected return equation and compute its value (G). What mathematical and operational effect does setting $\gamma < 1$ have on later rewards?

Solution

Computation:
$G = R_0 + \gamma R_1$[cite: 9]
$G = 2 + 0.5(6) = 2 + 3 = \mathbf{5}$[cite: 9]

Effect: A discount factor below 1 reduces the weight of later rewards[cite: 9]. Operationally, this causes the agent to prefer immediate payoffs over delayed gratification.

Part 2: Student Independent Practice Solutions

A. Interpreting Action-Values (Q-Values)

Scenario: A cyber-defense agent is actively monitoring a network node under high load. The agent calculates the following action values for its current state (s) based on its current security policy (π):

- $Q^{\pi}(s, \text{Block}) = 8$
- $Q^{\pi}(s, \text{Monitor}) = 5$
- $Q^{\pi}(s, \text{Ignore}) = 2$

1. **Question:** If the agent utilizes a **greedy choice**, which action will it select?

   Solution

   The agent will select the **Block** action (since 8 is the highest available Q-value)[cite: 9].

2. **Question:** What exactly does the largest Q-value (8) represent in the context of this MDP?

   Solution

   The value 8 is the expected discounted return from taking the Block action in state s and then strictly following policy π for all subsequent actions[cite: 9].

B. Comparing Returns

Scenario: An Autonomous Ground Vehicle (AGV) must clear a path through a contested urban environment. It has two options:

- **Action A (Direct Route):** Gives an immediate reward of +4 and successfully ends the route.
- **Action B (Covert Route):** Gives a reward of +1 now, and a reward of +4 on the next time step.

Given a discount factor of $\gamma = 0.5$:

1. **Question:** Compute the expected return for Action A.

   Solution

   $G_A = \mathbf{4}$ (because the action ends immediately, there are no future rewards to discount)[cite: 10].

2. **Question:** Compute the expected return for Action B.

   Solution

   $G_B = 1 + 0.5(4)$[cite: 10]
   $G_B = 1 + 2 = \mathbf{3}$[cite: 10]

3. **Question:** Compare the returns and identify the preferred action for the AGV to take.

Solution

Because the expected return of A (4) is mathematically greater than the expected return of B (3), the preferred action is **Action A**[cite: 10].